Name → Ravi

GitHub → github/ravi3028/assignment3-cloud computing.git

What I learnt →

How sequence-level knowledge works training a small student (python-tiny) on a teacher's (phi-3.5) generated text, not its weights; logit level was not possible as ollama return only text.

g also learnt object-storage (reduced FS/S3), running kubernetes with k3d (pods, resources, limits, HPA, scaling upto 6), stateless NACL rule ordering, and (I with a quality gate.

Challenges and solutions →

→ Teacher added prose after its JSON (17% valid) → used ollama format: "json". 100% valid.

→ Student stored 0% → a trailing space broke tokenization; ending it as "json". fixed it (0 → 99%)

→ my laptop could not run docker (no WSL2 / virtualization) → ran the K3d cluster on Github Actions

→ kept the image ≤ 2GB with CPU only Pytorch, weights as release assets, not in git

findings →

Student fall from 4% vs teachers 48%, 99.1% valid JSON a 7cm model underfits, cluster: Image 1.74 GD / cold start 15, p95 3300ms at 2.41 rps, load balanced 51/49, SLO 99.1% / −0.91 JNR per 100 requests / NDCG 6/6.

Improvements →

Requests batching, more epochs / a larger students, a shared work queues, a multi-node cluster with a Pod PIS auth in budget and per-service scoped S3 keys.